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Grade	**70.00** out of 150.00 (**47**%)

Question 1

Partially correct

Mark 5.00 out of 6.00

MoK Neurosciences CXEMI010

Prefrontal syndromes

A 65-year-old man has presented to the clinic with frontal lobe deficits. Identify two clinical features commonly seen in each type of pre-frontal syndrome

1. Medial prefrontal syndrome — Paucity of spontaneous behaviour ✔

Poverty of speech ✔

2. Orbitofrontal syndrome — Socially appropriate behaviour ✖

Explosive outbursts ✔

3. Dorsolateral prefrontal syndrome — Diminished planning ✔

Executive dysfunction ✔

Explanation

Medial (ventromedial) prefrontal syndrome, also called apathetic, or pseudo depressive type is featured by a paucity of spontaneous behaviours, sparse verbal output, reduced social knowledge and judgement of harmful intent.

Orbitofrontal syndrome is also called disinhibited or pseudopsychopathic type and is characterised by poor impulse control, explosive outbursts, and inappropriate behaviour.

Dorsolateral is called as dysexecutive or disorganized type is featured by cognitive dysfunction, diminished judgment, planning and poor insight. The patients are also described to be concrete and inflexible in nature.

The correct answer is:

Medial prefrontal syndrome - Poverty of speech, Paucity of spontaneous behaviour

Orbitofrontal syndrome - Poor impulse control, Explosive outbursts

Dorsolateral prefrontal syndrome - Executive dysfunction, Diminished planning

Question 2

Correct

Mark 3.00 out of 3.00

Neuropsychological measurement

Identify the test which could be used in each of the following situations

Which of the above tests use the following question: 'How many camels are there in England?' (Choose one)	Cognitive Estimation Test ✔
Scores are given for pronouncing the words DRACHM, SUPERFLUOUS and PLACEBO correctly (Choose one)	National Adult Reading Test ✔
The subject is asked to connect scattered numbers and alphabets in the sequence 1A2B3C4D5E6F..., using a pen and paper. (Choose one)	Trail Making Test B ✔

Your answer is correct.

In cognitive estimates test, factual questions that require abstract processing are asked. This is a test of frontal function.

In NART, scores are given for correct pronunciation of complex phonetic words.

In Trail Making Test A, the subject has to simply connect 1-2-3-4-5-6...etc as a sequence. In Trail Making Test B, the subject has to connect the sequence 1A2B3C4D5E6F...etc.

The correct answer is: Which of the above tests use the following question: 'How many camels are there in England?' (Choose one) → Cognitive Estimation Test, Scores are given for pronouncing the words DRACHM, SUPERFLUOUS and PLACEBO correctly (Choose one) → National Adult Reading Test, The subject is asked to connect scattered numbers and alphabets in the sequence 1A2B3C4D5E6F..., using a pen and paper. (Choose one) → Trail Making Test B

Question 3

Partially correct

Mark 1.00 out of 4.00

ECG
Match the following ECG changes with the most likely scenario.

A 50-year-old woman has been feeling low in mood, and finding it difficult to sleep. She is prescribed a medication by her GP, which is increased over a period of 4 weeks. Subsequently she has an ECG after reporting palpitations and dizziness.	Prolongation of QTc interval ✖
A young man with first episode psychosis is prescribed a medication and three months later has a routine ECG.	QTc prolongation of >510ms ✖
A severely depressed man is prescribed a medication, which leaves him with a dry mouth. A few days later he is found at home collapsed, having taken a significant overdose.	Torsades de Pointes with use of antidepressant ✖
A middle-aged man admitted to the ward with chronic psychotic illness reports some chest discomfort. An ECG is performed.	Saddle-shaped ST elevation ✔

Your answer is partially correct.

You have correctly selected 1.

1) Amitriptyline is an antidepressant medication that causes dry mouth and QTc prolongation leading to severe ventricular arrhythmias, collapse and death.

2) At higher doses, antidepressants may cause Torsades de Pointes.

3) Prolongation of QTc interval is a common side effect of antipsychotic medication.

4) This is clozapine-induced pericarditis.

The correct answer is: A 50-year-old woman has been feeling low in mood, and finding it difficult to sleep. She is prescribed a medication by her GP, which is increased over a period of 4 weeks. Subsequently she has an ECG after reporting palpitations and dizziness. → Torsades de Pointes with use of antidepressant, A young man with first episode psychosis is prescribed a medication and three months later has a routine ECG. → Prolongation of QTc interval, A severely depressed man is prescribed a medication, which leaves him with a dry mouth. A few days later he is found at home collapsed, having taken a significant overdose. → QTc prolongation of >510ms, A middle-aged man admitted to the ward with chronic psychotic illness reports some chest discomfort. An ECG is performed. → Saddle-shaped ST elevation

Question 4

Correct

Mark 3.00 out of 3.00

Sequelae of Eating Disorders

Which of the following sequelae of eating disordered behaviour would be associated with the below descriptions?

1. Miss Jane Mellors is a 20-year-old woman with a history of bulimia nervosa. She stopped purging via self-induced vomiting 2 days ago and now describes painless swellings on her face. Parotitis ✔

2. Miss Clare Phillips is a 19-year woman with a history of anorexia nervosa complicated by laxative misuse. She has recently stopped using laxatives and is currently engaging more with the planned treatment. She informs her care co-ordinator that she has been experiencing abdominal swelling. Decrease in intestinal motility ✔

3. Miss Chantelle Carmichael is a 32-year-old woman presenting to the emergency department with severe abdominal pain and vomiting. She describes a sharp epigastric pain radiating to the back. She claims that this pain gets worse with eating. Her partner refuses to believe her as she has an ongoing history of eating disorder with purging behaviours. Pancreatitis ✔

Explanation: 1) Parotitis can be associated with bulimia nervosa, particularly after an individual ceases to purge. It is usually painless and resolves spontaneously. Medication to stimulate salivation may be useful.

2) Long-term laxative misuse is associated with overstimulation of the colon leading to constipation once the laxatives are discontinued. Ideally, laxatives would be gradually withdrawn to allow the colon to return to a normal tone in a graded fashion.

3) Sharp epigastric pain radiating to the back that gets exacerbated by eating is a feature of acute pancreatitis. The onset of acute pancreatitis in the absence of heavy alcohol use or biliary disorders should prompt an enquiry regarding eating disorders.

Correct Answer is :

1. Miss Jane Mellors is a 20-year-old woman with a history of bulimia nervosa. She stopped purging via self-induced vomiting 2 days ago and now describes painless swellings on her face. - Parotitis

2. Miss Clare Phillips is a 19-year woman with a history of anorexia nervosa complicated by laxative misuse. She has recently stopped using laxatives and is currently engaging more with the planned treatment. She informs her care co-ordinator that she has been experiencing abdominal swelling. - Decrease in intestinal motility

3. Miss Chantelle Carmichael is a 32-year-old woman presenting to the emergency department with severe abdominal pain and vomiting. She describes a sharp epigastric pain radiating to the back. She claims that this pain gets worse with eating. Her partner refuses to believe her as she has an ongoing history of eating disorder with purging behaviours. - Pancreatitis

Question 5

Partially correct

Mark 1.00 out of 3.00

Neurological signs
Which of the following neurological signs are associated with the below scenarios?

1. Mr William Wright is a 70-year-old former electrician who has suffered a left hemisphere CVA. He has subsequently recovered his right arm and leg function completely. He has discovered that, although he has suffered no residual weakness or loss of sensation, he cannot not use his tools as usual. If told to do something he cannot carry out the action but can do so if he is not thinking consciously about it.

Ideational apraxia ✖

2. Mrs Lesley Nielson is a 68-year-old woman who has suffered a CVA. She now finds it difficult to perform multi-stage actions, such as making a cup of tea, even though she can understand what to do and can repeat single actions if she is shown them by others.

Ideomotor apraxia ✖

3. Mr Liam Feeley is a 58-year-old man who has suffered a significant head injury in a road traffic collision. He now finds that he cannot identify landmarks near his home, despite having been an avid local historian. Visuospatial agnosia ✔

Explanation:

1) This is ideomotor apraxia. These patients have deficits in their ability to plan or complete motor actions that rely on semantic memory. There is inability to convert an idea into action. It is usually caused by a dominant frontal or parietal lobe lesion. They may not be able to pick up a phone when asked to do so, but can perform the action without thinking when the phone rings.

2) This is ideational apraxia. Patients have an inability to conceptualise a task and impaired ability to complete multistep actions. They struggle to select and carry out an appropriate motor program. For example, the patient may complete actions in incorrect orders, such as buttering bread before putting it in the toaster, or putting on shoes before putting on socks.

3) Visuospatial agnosia is a loss of the sense of "whereness" in the relation of oneself to one's environment and in the relation of objects to each other.

Correct Answer is :

1. Mr William Wright is a 70-year-old former electrician who has suffered a left hemisphere CVA. He has subsequently recovered his right arm and leg function completely. He has discovered that, although he has suffered no residual weakness or loss of sensation, he cannot not use his tools as usual. If told to do something he cannot carry out the action but can do so if he is not thinking consciously about it. - Ideomotor apraxia

2. Mrs Lesley Nielson is a 68-year-old woman who has suffered a CVA. She now finds it difficult to perform multi-stage actions, such as making a cup of tea, even though she can understand what to do and can repeat single actions if she is shown them by others. - Ideational apraxia

3. Mr Liam Feeley is a 58-year-old man who has suffered a significant head injury in a road traffic collision. He now finds that he cannot identify landmarks near his home, despite having been an avid local historian. - Visuospatial agnosia

Question 6

Partially correct

Mark 5.00 out of 6.00

Genetic Loci

Regarding the following conditions, please identify the specific loci associated with the following diseases:

1. Parkinson's Disease - Choose TWO options
   - PRKN gene located on chromosome 6 ✔
   - SNCA gene located on chromosome 4 ✔
2. Fronto-temporal Dementia - Choose THREE options
   - MAPT gene located on chromosome 17 ✔
   - GRN gene located on chromosome 17 ✔
   - C9ORF72 gene located on chromosome 17 ✖
3. Huntington Disease - Choose ONE option
   - HTT gene located on chromosome 4 ✔

Explanation:

1) Familial cases of Parkinson disease can be caused by mutations in the LRRK2, PARK7, PINK1, PRKN, or SNCA gene.

- The alpha-synuclein (SNCA) gene located in the long arm of chromosome 4
- The parkin (PRKN) gene located in the long arm of chromosome 6.

2) Most familial frontotemporal dementia is caused by mutations in three genes. These are a recently discovered gene called C9ORF72, and genes for the proteins tau (MAPT) and progranulin (GRN).

- C9ORF72 is located at 9p21.2, which is the short (p) arm of chromosome 9 at position 21.2 and is also associated with Amyotrophic Lateral Sclerosis.

- MAPT is located at 17q21.31, which is the long (q) arm of chromosome 17 at position 21.31 and is associated with frontotemporal dementia with Parkinson-17 and progressive supranuclear palsy.

- GRN is located at 17q21.31, which is the long (q) arm of chromosome 17 at position 21.31.

3) Mutations (CAG trinucleotide repeats) in the HTT gene cause Huntington disease. It is located at 4p16.3, which is the short (p) arm of chromosome 4 at position 16.3. The HTT gene provides instructions for making a protein called huntingtin. (Ref: NIH. Genetics Home Reference [Internet]. U.S. National Library of Medicine. National Institutes of Health; Various Dates (Pages on Frontotemporal Dementia, Parkinson's and Huntington's) https://ghr.nlm.nih.gov/)

Correct Answer is :

1. Parkinson's Disease - Choose TWO options - SNCA gene located on chromosome 4, PRKN gene located on chromosome 6

2. Fronto-temporal Dementia - Choose THREE options - C9ORF72 gene located on chromosome 9, MAPT gene located on chromosome 17, GRN gene located on chromosome 17

3. Huntington Disease - Choose ONE option - HTT gene located on chromosome 4

Question 7

Partially correct

Mark 2.00 out of 3.00

Pathophysiology
Which of the following are associated with the below scenarios?

1. Mr Adam Lawn is a 37-year-old man with a history of alcohol dependence. His family have brought him into A&E after visiting him at home and finding that his memory appeared to be impaired. They reported that he kept forgetting that they were in his house.

Mamillary body alpha ketoglutarate ✔

2. Miss Kay Hughes is a 28-year-old woman with a diagnosis of depression who has been started on antidepressants. She wants to know about serotonin metabolism and asks about the main storage site for serotonin in the brain. Raphe nuclei ✔

3. Mrs Loretta Lovelace is a 22-year-old woman presenting with a history of recurrent hypersomnia associated with symptoms of hyperphagia, hypersexuality, and cognitive impairment. Cingulate gyrus ✖

Explanation:

1) Korsakoff's syndrome is due to thiamine deficiency, usually in the context of alcohol dependence. Thiamine is essential for the decarboxylation of pyruvate, and deficiency during this metabolic process is thought to cause damage to the medial thalamus and mammillary bodies of the posterior hypothalamus, as well as generalised cerebral atrophy.

2) The neurons of the raphe nuclei are the principal source of serotonin release in the brain.

3) Kleine-Levin syndrome is a recurrent hypersomnia associated with symptoms of hyperphagia, hypersexuality, and cognitive impairment. Kleine-Levin syndrome is rare, affecting an estimated 1-5 per million individuals. There are only approximately 200 reported cases to date in the literature. The symptoms of Kleine-Levin syndrome are characterized by their intermittent and periodic nature. During episodes, patients complain of excessive daytime sleepiness, despite sleeping anywhere from 12 to 21 hours per day. While awake, patients are often apathetic and report impairment in communication, concentration, and memory. The classic triad of hypersomnia, hyperphagia, and hypersexuality is not always present. In fact, in one large case series, only 45% of patients presented with all three symptoms. Hypersexuality is much more common in men than women. Thalamic hypoperfusion has been the most consistent finding in these studies when patients are examined during their symptomatic period. Once patients revert to their normal behavior, the thalamic hypoperfusion seems to resolve, while residual hypoperfusion persists in other brain locations. Fludeoxyglucose-positron emission tomography scanning has demonstrated asymmetric hypometabolism in the thalamus and hypothalamus.

Correct Answer is :

1. Mr Adam Lawn is a 37-year-old man with a history of alcohol dependence. His family have brought him into A&E after visiting him at home and finding that his memory appeared to be impaired. They reported that he kept forgetting that they were in his house. - Mamillary body alpha ketoglutarate

2. Miss Kay Hughes is a 28-year-old woman with a diagnosis of depression who has been started on antidepressants. She wants to know about serotonin metabolism and asks about the main storage site for serotonin in the brain. - Raphe nuclei

3. Mrs Loretta Lovelace is a 22-year-old woman presenting with a history of recurrent hypersomnia associated with symptoms of hyperphagia, hypersexuality, and cognitive impairment. - Hypothalamus

Question 8

Partially correct

Mark 4.00 out of 5.00

Developmental milestones
Identify the age of developmental milestones to achieve for each of the following:

1. Ability to build a tower of 6 cubes 2 years ✔

2. Crawling 6-9 months ✔

3. Copying a square 4 years ✔

4. Ability to say "dada" and "mama" correctly 12 months ✔

5. Ability to use simple sentences 4 years ✖

Explanation:

1) In terms of using blocks, children can usually stack 2 blocks by 15 months, 3 blocks by 18 months, 6-7 blocks by the age of 2 years and 9-10 blocks by the age of 3 years.

2) Children usually start to crawl at the age of 6-9 months. By 9-12 months children are "cruising" on furniture, walking with assistance and eventually walking independently, albeit unsteadily.

3) Children can usually copy circles by the age of 3 years, squares by the age of 4 and triangles from the age of 5-6 years.

4) At age of 12 months children can usually say a few single simple words, such as "mama" or "dada". By 2 years they can use around 50 words and use word pairings such as "car go" or similar.

5) By the age of 3 children can use simple sentences.

Correct Answer is :

1. Ability to build a tower of 6 cubes. - 2 years

2. Crawling - 6-9 months

3. Copying a square - 4 years

4. Ability to say "dada" and "mama" correctly. - 12 months

5. Ability to use simple sentences - 3 years

Question 9

Partially correct

Mark 5.00 out of 7.00

Mechanism of Action of Drugs for Dementia
Regarding the following anti-dementia medications:

1. What is Memantine's mechanism of action? Choose TWO options
   - Non-Competitive NMDA receptor antagonist ✔
   - 5-HT3 receptor antagonist ✔
2. What is Galantamine's mechanism of action? Choose TWO options
   - Nicotinic receptor modulator ✔
   - Selective and reversible acetylcholinesterase inhibitor ✔
3. What is Rivastigmine's mechanism of action? Choose TWO options
   - Butyrylcholinesterase inhibitor ✔
   - Selective and reversible acetylcholinesterase inhibitor ✖
4. What is Donepezil's mechanism of action? Choose ONE option
   - Pseudo-irreversible acetylcholinesterase inhibitor ✖

Explanation:

1) Memantine is a non-competitive NMDA receptor antagonist and also a 5-HT3 receptor antagonist.

2) Galantamine is a selective and reversible acetylcholinesterase inhibitor and also acts as a nicotine modulator.

3) Rivastigmine is a pseudo-irreversible acetylcholinesterase inhibitor and also acts as a butyrylcholinesterase inhibitor.

4) Donepezil is a selective and reversible acetylcholinesterase inhibitor. (Explanation: Taylor, D., Paton, C. and Kapur, S. (n.d.). The Maudsley prescribing guidelines in psychiatry. 12th ed. Chichester: John Wiley & Sons, pp.488)

Correct Answer is :

1. What is Memantine's mechanism of action? Choose TWO options - Non-Competitive NMDA receptor antagonist, 5-HT3 receptor antagonist

2. What is Galantamine's mechanism of action? Choose TWO options - Nicotinic receptor modulator, Selective and reversible acetylcholinesterase inhibitor

3. What is Rivastigmine's mechanism of action? Choose TWO options - Pseudo-irreversible acetylcholinesterase inhibitor, Butyrylcholinesterase inhibitor

4. What is Donepezil's mechanism of action? Choose ONE option - Selective and reversible acetylcholinesterase inhibitor

Question 10

Correct

Mark 5.00 out of 5.00

Switching antidepressants
What is the advised route for the following changes in medication?

Trazodone to Mirtazapine	Cross taper with caution ✔
Fluoxetine to Tranylcypromine	Withdraw and wait for 5-6 weeks ✔
Phenelzine to Sertraline	Withdraw and wait for 2 weeks ✔
Amitryptiline to citalopram	Half dose, start new drug, then slow withdrawal ✔
Fluoxetine to duloxetine	Stop and start other drug immediately ✔

Your answer is correct.

When swapping from one antidepressant to another, abrupt withdrawal should usually be avoided. Cross tapering is preferred, where the dose of the ineffective or poorly tolerated drug is slowly reduced while the new drug is slowly introduced. 1) Cross tape

The correct answer is: Trazodone to Mirtazapine → Cross taper with caution, Fluoxetine to Tranylcypromine → Withdraw and wait for 5-6 weeks, Phenelzine to Sertraline → Withdraw and wait for 2 weeks, Amitryptiline to citalopram → Half dose, start new drug, then slow withdrawal, Fluoxetine to duloxetine → Stop and start other drug immediately

Question 11

Correct

Mark 1.00 out of 1.00

Which one of the following is not a frontal lobe function test?

Select one:

- Tower of London
- Verbal fluency
- Luria's motor tests.
- California Verbal Learning Test ✔
- Cognitive estimates

Your answer is correct.

Frontal lobe tests include Stroop test, Halstead trial making, cognitive estimates, Tower of London, Wisconsin card sorting test, verbal fluency, Luria's motor tests and multiple errands tests. The California Verbal Learning Test (CVLT) is useful to assess verbal memory. Though it may be affected in patients with frontal lesions, CVLT is not a specific frontal lobe test.

The correct answer is: California Verbal Learning Test

Question 12

Correct

Mark 1.00 out of 1.00

Aura is a commonly occurring phenomenon seen in which type of seizures?

Select one:

- Grandmal seizures
- Simple partial seizures
- Complex partial seizures ✔
- Petitmal seizures
- Myoclonic seizures

Your answer is correct.

Auras are particularly common in temporal lobe complex partial epilepsy, generally lasting only a few seconds. Auras are typically stereotyped, e.g., autonomic or visceral aura (epigastric sensations) and may consist of depersonalization, derealisation and affective changes (like anxiety, euphoria), perceptual experiences like illusions and hallucinations.

The correct answer is: Complex partial seizures

Question 13

Correct

Mark 1.00 out of 1.00

In a Broca's area lesion, which of the following features is seen?

Select one:

- Naming is preserved
- Comprehension is preserved ✔
- Inflection is preserved
- Repetition is preserved
- Fluency is preserved

Your answer is correct.

Patients with Broca's aphasia produce slow, halting speech that is rarely grammatical. Typical Broca's aphasics eliminate inflections such as -ed and words not central to the meaning of the sentence, such as 'the' and 'and'.

The correct answer is: Comprehension is preserved

Question 14

Incorrect

Mark 0.00 out of 1.00

A 50-year-old woman began complaining of double vision and blurry vision 3 months ago and has since had diminishing interaction with her family, a paucity of thought and expression, and unsteadiness of gait. Her whole body appears to jump in the presence of a loud noise. An MRI scan and routine CSF examination are unremarkable. What is the most appropriate diagnosis?

Select one:

- Creutzfeldt-Jakob disease
- Alzheimer's disease
- Normal-pressure hydrocephalus ✖
- Parkinson's disease
- Transient global amnesia

Your answer is incorrect.

The neurological complaints occurring early in the course of Creutzfeldt-Jakob disease are often cerebellar or visual. Patients may complain of ataxia, clumsiness, or dysarthria, as well as diplopia, distorted vision, blurred vision, field defects, changes in colour perception, and visual agnosia. Ultimately cortical blindness may occur.

The correct answer is: Creutzfeldt-Jakob disease

Question 15

Correct

Mark 1.00 out of 1.00

George is a 5-year-old boy with poor social interactions and stereotypical behaviours. Which part of the brain would you expect to be dysfunctional?

Select one:

- Broca's area
- Cerebrum
- Basal ganglia
- Prefrontal cortex
- Vermis of cerebellum ✔

Your answer is correct.

This boy has features of autistic spectrum disorder that includes poor social interactions, and stereotypical behaviours. Hypoplasia of cerebellar vermis (involving to some extent the cerebellar hemispheres) is well-documented in autism. Purkinje cell count in the cerebellum is significantly lower in patients with autism.

The correct answer is: Vermis of cerebellum

Question 16

Correct

Mark 1.00 out of 1.00

Which of the following structures are employed in processing emotionally salient memories?

Select one:

- Basal ganglia
- Angular gyrus
- Amygdala ✔
- Cerebellum
- Prefrontal cortex

Your answer is correct.

The amygdala and hippocampal complex, two medial temporal lobe structures, are linked to two independent memory systems, each with unique characteristic functions. In emotional situations, these two systems interact in subtle but important ways. Specifically, the amygdala can modulate both the encoding and the storage of hippocampal-dependent memories. The hippocampal complex, by forming episodic representations of the emotional significance and interpretation of events, can influence the amygdala response when emotional stimuli are encountered. Ref: Phelps EA. Human emotion and memory: interactions of the amygdala and hippocampal complex. Current opinion in neurobiology. 2004 Apr 30;14(2):198-202.

The correct answer is: Amygdala

Question 17

Correct

Mark 1.00 out of 1.00

A gentleman presents with a 1-day history of sudden onset symptoms of akinetic mutism. What is the vascular origin of this presentation?

Select one:

- Basilar artery
- Anterior cerebral artery ✔
- Posterior inferior cerebellar artery
- Posterior cerebral artery
- Middle cerebral artery

Your answer is correct.

Bilateral infarct involving the anterior cerebral artery produces quadriparesis (legs weaker than arms) and akinetic mutism (ventromedial or cingulate syndrome). Ref: Freemon FR. Akinetic mutism and bilateral anterior cerebral artery occlusion. Journal of Neurology, Neurosurgery & Psychiatry. 1971 Dec 1;34(6):693-8.

The correct answer is: Anterior cerebral artery

Question 18

Correct

Mark 1.00 out of 1.00

Which of the following brain regions is associated with thermoregulation?

Select one:

- Pontine nucleus
- Prefrontal cortex
- Pre-optic area ✔
- Ventral striatum
- Medial temporal lobe

Your answer is correct.

The pre-optic area is a region of the hypothalamus. It is responsible for thermoregulation. It receives nervous stimulation from receptors in the skin, mucous membranes and hypothalamus Ref: Boulant JA. Role of the preoptic-anterior hypothalamus in thermoregulation and fever. Clinical infectious diseases. 2000 Oct 1;31(Supplement 5):S157-61.

The correct answer is: Pre-optic area

Question 19

Correct

Mark 1.00 out of 1.00

John is a 67-year old man admitted to the geriatric ward with a history of recent weight loss. On obtaining further history, he admits to loss of the sense of smell. This feature is suggestive of

Select one:

- Cerebellar pathology
- Occipital lobe pathology
- Frontal lobe pathology ✔
- Parietal lobe pathology
- Temporal lobe pathology

Your answer is correct.

Anosmia is the inability to perceive odour or loss of the sense of smell. A permanent loss of smell may be caused either by a loss of olfactory receptor neurons in the nose or a damage to the olfactory nerve or the brain areas that process smell, usually after head injury or due to tumors of the frontal lobe.

The correct answer is: Frontal lobe pathology

Question 20

Correct

Mark 1.00 out of 1.00

Which of the following is the precursor of serotonin?

Select one:

- 5-hydroxydopamine
- 5-hydroxyindoleacetic acid
- 5-hydroxytryptophan ✔
- 5-hydroxytryptamine
- Anandamide

Your answer is correct.

Note that 5-hydroxytryptamine is serotonin itself.

The correct answer is: 5-hydroxytryptophan

Question 21

Correct

Mark 1.00 out of 1.00

The somatodendritic inhibition of 5-HT release is regulated by

Select one:

- 5-HT1A receptors ✔
- 5-HT3 receptors
- 5-HT4 receptors
- 5-HT5 receptors
- 5-HT2 receptors

Your answer is correct.

The 5-HT1A receptor is found at both somatodendritic and post-synaptic sites. Somatodendritic 5-HT1A receptors in the dorsal and median raphe nuclei are autoreceptors regulating 5-HT neuronal firing. Post-synaptic 5-HT1A receptors are found in the hippocampus and periaqueductal grey matter where they regulate behaviours such as resilience, impulsivity and restraint of excessive response to stress. The 5-HT1A receptor may be an important target in the action of antidepressants. (Ref: Fundamentals of clinical psychopharmacology- 18)

The correct answer is: 5-HT1A receptors

Question 22

Correct

Mark 1.00 out of 1.00

A 70-year-old woman awakens with inability to walk and is found on the floor by her family. Her examination shows right face, arm, and leg weakness and numbness. What is the most likely cause of her gait problems?

Select one:

- Parkinsonism
- A right hemisphere infarction
- A right cerebellar infarction
- A left hemisphere infarction ✔
- Drug toxicity

Your answer is correct.

The presentation is suggestive of a cerebral infarct, most likely to be on the left side as the woman is presenting with right-sided weakness.

The correct answer is: A left hemisphere infarction

Question 23

Correct

Mark 1.00 out of 1.00

Mr. P is a 77-year-old gentleman suffering from gradually progressive cognitive decline for last 6 months. He has history of Parkinson's disease for last 3 years and suffers from visual hallucinations. The chemical that is mainly involved and deficient in this condition is

Select one:

- Glutamate
- Acetylcholine ✔
- Noradenaline
- GABA
- Serotonin

Your answer is correct.

The diagnosis is Parkinson's disease dementia. In both Lewy body dementia and Parkinson's disease dementia, brain acetylcholine levels are reduced similar to Alzheimer's disease. Therefore acetylcholinesterases (Anti-dementia drugs) could be used in the treatment of Lewy body dementia.

The correct answer is: Acetylcholine

Question 24

Correct

Mark 1.00 out of 1.00

Mr. Y was diagnosed with HIV almost 2 years ago. He is now presenting with cognitive impairment. Which of the following cells is least likely to be affected?

Select one:

- Lymphocytes
- None of the above
- Neurones ✔
- Macrophages
- Microglia

Your answer is correct.

Theoretically all the main cell types of the CNS, astrocytes, oligodendrocytes, neurons, perivascular macrophage and microglia, can be infected by HIV-1 since they possess the receptors and/or co-receptors for HIV-1 entry, but only the latter two are the most commonly infected cells by HIV-1.

Most studies have indicated an absence of in vivo infection in neurons. It is unclear whether detection of infected neurons is complicated by the loss of the infected neuronal populations.

(Excerpt from Ghafouri et al., Retrovirology 2006, 3:28).

The correct answer is: Neurones

Question 25

Correct

Mark 1.00 out of 1.00

Regional cerebral blood flow is affected in schizophrenia. Which of the following area is most likely to be affected?

Select one:

- Left medial temporal
- Prefrontal cortex ✔
- Raphe nuclei
- Basal ganglia
- Bilateral parietal

Your answer is correct.

Hypofrontality has been repeatedly demonstrated.

In schizophrenia:
Hypoperfusion: frontal lobes and frontal cortex, anterior and medial cingulate gyri, and parietal lobes.
Hyperperfusion: cerebellum, brainstem and thalamus.

The correct answer is: Prefrontal cortex

Question 26

Correct

Mark 1.00 out of 1.00

The post-mortem examination done on an elderly gentleman who suffered from a slowly progressive non-amnesic cognitive impairment, disinhibition and pill-rolling tremors reveals ubiquitin inclusions. Which one among the following genes is most likely to reveal a mutation?

Select one:

- Progranulin ✔
- Prion protein
- Neurexin
- Presenilin-2
- Dysbindin

Your answer is correct.

This is a case of frontotemporal dementia (FTD). FTD is caused by mutations in progranulin (PGRN) especially in familial FTD. PGRN is located on chromosome 17q21.31.

The correct answer is: Progranulin

Question 27

Correct

Mark 1.00 out of 1.00

The density of D2 receptors seen in children before 5 years of age is greater than adult levels. Their regression to adult levels is reached during the second decade. The developmental process through which this is achieved is called

Select one:

- Downregulation
- Pruning ✔
- Desensitisation
- Waning
- Habituation

Your answer is correct.

Synaptic 'pruning' refers to the natural elimination of unnecessary synapses that normally occurs in adolescence, making the brain more organised and efficient.

The correct answer is: Pruning

Question 28

Correct

Mark 1.00 out of 1.00

A disturbance in cerebral torque is proposed as the underlying pathological change in which of the following disorders?

Select one:

- OCD
- Autism
- Grand mal epilepsy
- PTSD
- Psychosis ✔

Your answer is correct.

'Cerebral torque' refers to opposing right-left asymmetries of frontal and parieto-occipital regions of the brain.

Crow's hypothesis and observations suggest that a primary disturbance in psychosis is in brain structure, and this is related to cerebral asymmetry in the form of torque.

The correct answer is: Psychosis

Question 29

Correct

Mark 1.00 out of 1.00

Which of the following is the most common opportunistic infection of CNS in patients with AIDS?

Select one:

- Cryptococcal meningitis
- Neurocysticercosis
- Toxoplasma gondii ✔
- Cytomegalovirus infection
- Tuberculous meningitis

Your answer is correct.

Toxoplasmic encephalitis (TE) is caused by the protozoan Toxoplasma gondii. The disease occurs almost exclusively because of reactivation of latent tissue cysts. Primary infection occasionally is associated with acute cerebral or disseminated disease. Seroprevalence varies substantially among different communities (e.g., approximately 15% in the United States and 50%--75% in certain European countries). Prior to the introduction of antiretroviral drugs, the 12-month incidence of TE was approximately 33%. The incidence has decreased substantially with the initiation of ART. Clinical disease is rare among patients with CD4+ T lymphocyte counts more than 200 cells/µL. Primary infection occurs after eating undercooked meat containing tissue cysts or ingestion of oocysts that have been shed in cat feces and have sporulated in the environment (which requires at least 24 hours). No transmission of the organism occurs by person-to-person contact (Retrieved from http://www.cdc.gov/mmwr/preview/mmwrhtml/rr58e324a1.htm).

The correct answer is: Toxoplasma gondii

Question 30

Correct

Mark 1.00 out of 1.00

Which of the following is a glucocorticoid receptor antagonist?

Select one:

- D-Cycloserine
- Clozapine
- Mifepristone ✔
- Agomelatine
- Gabapentin

Your answer is correct.

Mifepristone is a glucocorticoid receptor antagonist. It is a blocker of type 2 glucocorticoid and progestin receptors, and the preliminary research findings suggest that it could effectively ameliorate symptoms of major depression.

The correct answer is: Mifepristone

Question 31

Correct

Mark 1.00 out of 1.00

The EEG findings of a patient shows multiple spike and wave activity. Identify the disease.

Select one:

- Absence seizure
- Creutzfeldt Jacob disease
- Subacute sclerosing panencephalitis
- Herpes encephalitis
- Myoclonic epilepsy ✔

Your answer is correct.

Common EEG findings include rapid spike wave and poly spike wave.

The correct answer is: Myoclonic epilepsy

Question 32

Correct

Mark 1.00 out of 1.00

The substance that is secreted by hypothalamus and enhances food intake is

Select one:

- Neuropeptide Y ✔
- Beta endorphins
- Neurotensin
- Vasopressin
- Cholecystokinin

Your answer is correct.

Food intake is regulated by various factors such as neuropeptide Y. Neuropeptide Y potently induces an increase in food intake, and simultaneously stimulates arginine-vasopressin (AVP) secretion in the brain (Retrieved from http://www.ncbi.nlm.nih.gov/pubmed/19766110).

The correct answer is: Neuropeptide Y

Question 33

Correct

Mark 1.00 out of 1.00

There is increased concentration in axon terminal prior to depolarisation with which of the following ions?

Select one:

- Calcium
- Nitrogen
- Sodium
- Hydrogen
- Potassium ✔

Your answer is correct.

There is a negative resting membrane potential of around -70mV. It is maintained by the sodium pump, which actively transports sodium ions out of the cell and potassium ions into the cell. There is increased concentration of potassium ions in axon terminal prior to depolarization. An action potential is propagated by the depolarization spreading laterally to adjacent parts of the neurone. An action potential is initiated in the axon hillock when the synaptic signals received by the dendrites and soma are sufficient to raise the intracellular potential from -70 mV to the threshold potential of - 55mV. When this potential is reached, the Na+ channels present in the axon initial segment will open. This Na+ influx causes a rapid reversal of the membrane potential from -90 to +40 mV. When the membrane potential reaches +40mV, the Na+ channels close and the voltage-gated K+ channels open. K+ ions move out of the axon, and 'repolarizes' the membrane.

The correct answer is: Potassium

Question 34

Correct

Mark 1.00 out of 1.00

Which of the following is least likely to be correct with regard to REM latency and depression? (June 2009, Nov2009)

Select one:

- REM sleep latency reduction precedes onset of major depressive disorder
- Shortened REM latency is specific to depression ✔
- Relatives of depressed patients have increased prevalence of short REM latency
- Reduced REM latency could persist even after successful treatment of depressive episode
- REM latency reduction predicts higher relapse risk

Your answer is correct.

Short REM latency is associated with increased risk of major depression beyond the familial risk associated with a depressed proband. Consistent with other heritable traits, both REM latency reduction and Slow Wave Sleep abnormalities have been observed preceding a first episode of depression in patients. But it has low predictive value as substantial number of those who have short REM latency, do not develop depression. Short REM latency is not specific to depression; it may be seen in PTSD, schizophrenia, eating disorders, etc

The correct answer is: Shortened REM latency is specific to depression

Question 35

Correct

Mark 1.00 out of 1.00

Mr. P, a patient attending your clinic, was started on clozapine a few weeks ago. He now complains of shortness of breath, tachycardia and fever. Which of the following would you request urgently?

Select one:

- ECG
- Chest x-ray
- CPK
- D-dimer
- Troponin ✔

Your answer is correct.

The most important consideration is clozapine-induced myocarditis. It occurs within 6-8 weeks of starting clozapine (median 3 weeks). Symptoms include fever, tachycardia, flu-like symptoms, fatigue, dyspnea (with increased respiratory rate) and chest pain. One proposed monitoring protocol recommends obtaining baseline troponin I/T, C-reactive protein and echocardiography, and monitoring troponin and C-reactive protein on days 7, 14, 21 and 28. At least twice the upper limit of normal troponin was found in 90% of cases. In addition, persistent abnormally high heart rate or signs or symptoms consistent with infective illness should be followed by daily troponin and C-reactive protein investigation until features resolve. Ref: Ronaldson et al., Aust N Z J Psychiatry. 2011 Jun;45(6):458-65. http://www.ncbi.nlm.nih.gov/pubmed/21524186

The correct answer is: Troponin

Question 36

Correct

Mark 1.00 out of 1.00

Prolonged QTc interval is associated with:

Select one:

- Naloxone
- Diazepam
- Naltrexone
- Buprenorphine
- Methadone ✔

Your answer is correct.

Methadone alone or combined with other arrythmogenic agents may increase the likelihood of QT interval prolongation, which is associated with torsades de pointes and can be fatal. ECG monitoring is therefore recommended whilst taking methadone.

The correct answer is: Methadone

Question 37

Correct

Mark 1.00 out of 1.00

Sally has a diagnosis of schizophrenia and is currently taking risperidone tablets. Her prolactin level tested recently shows an elevation that is twice the value recorded prior to commencing the treatment. Which of the following drugs could be considered as the best possible add-on treatment?

Select one:

- Olanzapine
- Aripiprazole ✔
- Sulpride
- Quetiapine
- Amisulpride

Your answer is correct.

For most patients with symptomatic hyperprolactinaemia, a switch to non-prolactin elevating drug is the first choice. An alternative with modest but evolving evidence is to add aripiprazole to existing treatment. Hyperprolactinaemia and related symptoms are reported to improve fairly promptly following the addition of aripiprazole. Ref: Mir et al., Psychopharmacol. 2008 May;22(3):244-53.

The correct answer is: Aripiprazole

Question 38

Correct

Mark 1.00 out of 1.00

A 47-year-old man developed seizures on taking an antipsychotic medication. The psychiatrist treating him initially reduced the dose of his antipsychotic, and later started him on sodium valproate. Which of the following antipsychotic drugs is most likely to produce this effect?

Select one:

- Clozapine ✔
- Haloperidol
- Fluphenazine
- Pimozide
- Risperidone

Your answer is correct.

Seizures may occur at any time whilst taking clozapine. It is a dose and plasma level-related effect. Prophylactic valproate or lamotrigine is used in high-risk subjects when titrating to a high dose of clozapine. EEG abnormalities are common in those on clozapine treatment; these abnormalities do not always indicate seizure activity.

The correct answer is: Clozapine

Question 39

Correct

Mark 1.00 out of 1.00

Mr. P developed cardiomyopathy on taking this medication for the last 9 months. The most likely drug is

Select one:

- Clozapine ✔
- Amisulpride
- Olanzapine
- Risperidone
- Quetiapine

Your answer is correct.

Clozapine is associated with myocarditis and cardiomyopathy. Myocarditis seems to occur within 6-8 weeks of starting clozapine (median 3 weeks) and cardiomyopathy may occur later in treatment (median 9 months) but both may occur at anytime.

The correct answer is: Clozapine

Question 40

Correct

Mark 1.00 out of 1.00

Which one among the following is a risk factor for SSRI induced hyponatraemia?

Select one:

- Young age
- Male sex
- Cold weather
- Low body mass ✔
- High initial sodium

Your answer is correct.

Risk factors for SSRI induced hyponatraemia include old age, female sex, low body weight, low baseline sodium concentration, history of hyponatraemia, warm weather, reduced renal function, medical co-morbidity and some drug treatments like carbamazepine, antipsychotics, thiazide diuretics, NSAIDs, cancer chemotherapy and drugs like tramadol, omeprazole and trimethoprim.

The correct answer is: Low body mass

Question 41

Correct

Mark 1.00 out of 1.00

Sally has depression but would accept her treatment only if the probability of drug induced weight gain is low. Which of the following antidepressants is best to avoid?

Select one:

- Venlafaxine
- Mirtazapine ✔
- Fluoxetine
- Sertraline
- Duloxetine

Your answer is correct.

Mirtazapine is associated with weight gain in around 1/5th of subjects, especially in the first 4 weeks of treatment. This effect may be secondary to its effects on the 5-HT2C and H1 receptors. Ref: Laimer M, Kramer-Reinstadler K, Rauchenzauner M, et al. Effect of mirtazapine treatment on body composition and metabolism. J Clin Psychiatry. 2006;67:421-424

The correct answer is: Mirtazapine

Question 42

Correct

Mark 1.00 out of 1.00

Which of the following is primarily tested by using Abnormal Involuntary Movement Scale (AIMS)?

Select one:

- Akathisia
- Tardive dyskinesia ✔
- Psychomotor agitation
- Torticollis
- Jaw stiffness

Your answer is correct.

The AIMS records the occurrence of tardive dyskinesia (TD) in patients receiving neuroleptic medications. This test is used to detect TD and to follow the severity of a patient's TD over time.

The correct answer is: Tardive dyskinesia

Question 43

Correct

Mark 1.00 out of 1.00

Which one of the following is the least likely feature associated with Neuroleptic Malignant Syndrome?

Select one:

- Raised CPK levels
- Hyperthermia
- Autonomic lability
- Persistent bradycardia ✔
- Muscle rigidity

Your answer is correct.

Bradycardia is not a feature seen in patients with NMS. The symptoms and signs of NMS include hyperthermia, rigidity, agitation, altered levels of consciousness, tachycardia, tachypnoea, hypertension, tremor, diaphoresis, raised CPK levels, leukocytosis and metabolic acidosis.

The correct answer is: Persistent bradycardia

Question 44

Correct

Mark 1.00 out of 1.00

The upper normal value of QTc in females is

Select one:

- 400 msec
- 650 msec
- 470 msec ✔
- 440 msec
- 500 msec

Your answer is correct.

The upper normal value of QTc in males is 440 msec and females is 470 msec. Mounting evidence links QTc values over 500 msec to a clearly increased risk of arrhythmia.

The correct answer is: 470 msec

Question 45

Incorrect

Mark 0.00 out of 1.00

Which of the following drugs increases the incidence of cleft palate when used in pregnant women?

Select one:

- Topiramate ✖
- Sertraline
- Lamotrigine
- Gabapentin
- Lithium

Your answer is incorrect.

Lamotrigine increases the incidence of cleft palate. There are some anecdotal reports linking valproate and diazepam as well to this abnormality.

The correct answer is: Lamotrigine

Question 46

Correct

Mark 1.00 out of 1.00

Which class of antidepressants, when used during pregnancy increases the risk of cleft lip/cleft palate in the newborn?

Select one:

- Monoamine oxidase inhibitors
- Noradrenergic and specific serotonergic antidepressant
- Selective serotonin reuptake inhibitors ✔
- Tricyclic antidepressant
- Norepinephrine reuptake inhibitor

Your answer is correct.

Existing research evidence has shown a connection between prenatal use of SSRI anti-depressants and the subsequent formation of birth defects like cleft lip and palate. In 2006, a Danish population-based cohort study published in Epidemiology reported that women taking SSRI antidepressants from 30 days before conception to the end of the first trimester were more likely to give birth to babies with congenital formations, which may include cleft lip and palate.

The correct answer is: Selective serotonin reuptake inhibitors

Question 47

Incorrect

Mark 0.00 out of 1.00

Which of the following adverse effects has not been reported in association with therapeutic doses of SSRIs?

Select one:

- Hypertension
- Hallucinations
- Akathisia
- Galactorrhea ✖
- Photosensitivity

Your answer is incorrect.

Several published reports of SSRI-induced movement disorders are available; they mostly report akathisia, dyskinesia, parkinsonism and occasionally dystonia and bruxism. Most of these were reported when using normal therapeutic doses. SSRIs, especially escitalopram and paroxetine, are occasionally associated with galactorrhea. Though rare, photosensitivity and hallucinations have been anecdotally reported with SSRIs. Nevertheless, therapeutic range of SSRI dose is not associated with systemic hypertension, though SNRIs such as venlafaxine carries this risk especially above the dose of 225mg/day. Ref: Ann Pharmacother. 1998 Jun;32(6):692-8.; Clin Exp Dermatol. 2009 Dec;34(8):e763-5.; J Clin Psychopharmacol. 2000 Oct;20(5):579-80.; Br J Clin Pharmacol. 1997;44277–81.; Journal of Clinical Psychiatry. 1996 Oct 1;57(10):449-54.

The correct answer is: Hypertension

Question 48

Correct

Mark 1.00 out of 1.00

Which of these drugs is likely to cause tricuspid valve displacement towards the apex of the right ventricle in the newborn, when used during pregnancy?

Select one:

- Haloperidol
- Lithium ✔
- Valproate
- Clozapine
- Carbamazepine

Your answer is correct.

Ebstein's anomaly is a congenital heart defect in which the tricuspid valves are displaced towards the apex of the right ventricle. Lithium use has a well-known association with this cardiac malformation (relative risk is 10-20 times more than control). Some case reports have also implicated benzodiazepines, although this is rare.

The correct answer is: Lithium

Question 49

Incorrect

Mark 0.00 out of 1.00

The risk of developing seizures with the use of clozapine is increased by:

Select one:

- 0.5
- 0.4
- 0.6 ✖
- 0.2
- 0.1

Your answer is incorrect.

Patients can have generalized tonic-clonic seizures during treatment with clozapine. Life-table analysis predicts a cumulative 10% risk of seizures after 3.8 years of treatment. Clozapine-related seizures appear to be dose-related. High-dose therapy (greater than or equal to 600 mg/day) was associated with a greater risk of seizures (4.4%) than medium (300 to 600 mg/day; 2.7%) or low doses (less than 300 mg/day; 1.0%). Also, rapid upward titration may increase seizure risk. Ref: Devinsky O, Honigfeld G, Patin J. Clozapine-related seizures. Neurology. 1991 Mar 1;41(3):369)

The correct answer is: 0.1

Question 50

Correct

Mark 1.00 out of 1.00

Which of the following antidementia drugs undergo breakdown by cholinesterase itself with no hepatic involvement in metabolism?

Select one:

- Rivastigmine ✔
- Donepezil
- Galantamine
- Memantine
- Tacrine

Your answer is correct.

Rivastigmine undergoes no hepatic conversion.

The correct answer is: Rivastigmine

Question 51

Not answered

Marked out of 1.00

Phenelzine has been shown to be superior to placebo in the treatment of

Select one:

- Arachnophobia
- Zoophobia
- Agoraphobia
- Social phobia
- Obsessional symptoms

Your answer is incorrect.

The monoamine oxidase inhibitors (MAOIs) have shown superior efficacy to other antidepressants in the the treatment of generalized social phobia. Phenelzine in particular, has extensive placebo-controlled trial evidence in support of its use in social anxiety with nearly 67% patients showing significant improvement during acute treatment.

The correct answer is: Social phobia

Question 52

Not answered

Marked out of 1.00

Which one among the following statements about atypical antipsychotic drugs is incorrect?

Select one:

- Atypical antipsychotic drugs have relatively high serotonin HT-2A to dopamine D-2 receptor binding ratios
- The extrapyramidal safety of atypical antipsychotic drugs depends on the affinity for nigrostriatal dopamine D-2 receptors
- Relatively high 5HT-2A to dopamine D-2 binding ratios of atypical antipsychotic drugs may be responsible for the efficacy and safety advantages.
- Atypical antipsychotic drugs have less affinity for nigrostriatal than typical antipsychotic drugs
- The efficacy of antipsychotic drugs appear to be determined by the affinity for nigrostriatal dopamine receptors

Your answer is incorrect.

The efficacy of antipsychotic drugs appears to be determined by the affinity for mesolimbic dopamine receptors and not by the affinity for nigrostriatal receptors.

The correct answer is: The efficacy of antipsychotic drugs appear to be determined by the affinity for nigrostriatal dopamine receptors

Question 53

Not answered

Marked out of 1.00

Which of the following mood stabilisers is not recommended for prophylaxis of bipolar affective disorder?

Select one:

- Lithium
- Carbamazepine
- Topiramate
- Valproate
- Lamotrigine

Your answer is incorrect.

There are no large trials indicating the usefulness of topiramate in BPAD. Current evidence is sparse and limited.

The correct answer is: Topiramate

Question 54

Not answered

Marked out of 1.00

Which of the following is a prodrug whose metabolite is used as an individual compound therapeutically?

Select one:

- Citalopram
- Zopiclone
- Fluoxetine
- Risperidone
- Haloperidol

Your answer is incorrect.

Paliperidone is 9-OH-risperidone.

Citalopram is not a prodrug - escitalopram is the therapeutically active S-enantiomer of citalopram.

The correct answer is: Risperidone

Question 55

Not answered

Marked out of 1.00

Which antipsychotic has a longer therapeutic effect attributable to its active metabolite?

Select one:

- Amisulpride
- Risperidone
- Chlorpromazine
- Haloperidol
- Sulpride

Your answer is incorrect.

The apparent half-life of risperidone is 3 hours while the apparent half-life of 9-hydroxyrisperidone (also known as paliperidone) is about 21 hours (in extensive metabolizers). The pharmacokinetics of risperidone and 9-hydroxyrisperidone combined, after single and multiple doses, were similar in extensive and poor metabolizers, with an overall mean elimination half-life of about 20 hours (RISPERDAL M-TAB - Food and Drug Administration, http://www.fda.gov/ohrms/dockets/dockets/06p0245/06P-0245-EC1-Attach-1.pdf).

The correct answer is: Risperidone

Question 56

Not answered

Marked out of 1.00

Which of the following measures can help reduce lithium-induced tremors?

Select one:

- Drug holidays
- Single large dose
- Taking lithium after food
- Smaller, more frequent doses
- Change time of administration to mornings only

Your answer is incorrect.

Administer the lithium in smaller, more frequent doses. Propranolol (30 to 160 mg/d) may also be helpful.

The correct answer is: Smaller, more frequent doses

Question 57

Not answered

Marked out of 1.00

A psychiatric trainee has started lamotrigine in addition to a second mood stabiliser for a lady with a relapse of bipolar affective disorder. The pharmacist is concerned that the second mood stabiliser will increase possibility of toxicity of the lamotrigine. The second mood stabiliser is most likely to be

Select one:

- Olanzapine
- Sodium valproate
- Lithium
- Risperidone
- Carbamazepine

Your answer is incorrect.

Sodium valproate can inhibit the breakdown of lamotrigine and so can result in higher plasma levels of lamotrigine. Valproate also increases the risk of allergic rash seen with lamotrigine.

The correct answer is: Sodium valproate

Question 58

Not answered

Marked out of 1.00

Mr Julian Andrews is a 32-year-old man who has presented to his GP with symptoms of low mood, lack of enjoyment, poor sleep, low energy levels and diminished appetite. How long would he need to have these symptoms before he could be diagnosed with a depressive episode under ICD-10 criteria?

Select one:

- 14 days
- 7 days
- 28 days
- 6 months
- 2 months

Your answer is incorrect.

ICD-10 requires a person to have depressive symptoms most days, most of the time for at least 2 weeks before an episode of depression can be diagnosed.

The correct answer is: 14 days

Question 59

Not answered

Marked out of 1.00

Robin Wells is a 35 year old who has recently been diagnosed with rapid cycling bipolar affective disorder. Which of the following is true about this disorder?

Select one:

- Associated with stimulant use
- Associated with hyperthyroidism
- More common in males
- At least four episodes of bipolar disorder in 1 year
- At least two episodes of bipolar disorder in 1 year

Your answer is incorrect.

Rapid cycling bipolar affective disorder is defined as having 4 or more episodes of mood disturbance (manic, hypomanic, mixed or depressive) within 12 months.

The correct answer is: At least four episodes of bipolar disorder in 1 year

Question 60

Not answered

Marked out of 1.00

The Sally Anne Test is a psychological test used with children. Which of the following is TRUE regarding the Sally Anne Test?

Select one:

- It was devised by Piaget
- It is correctly interpreted by the age of 3
- Girls are more likely to correctly interpret the test than boys
- It is a test of second-order false belief tasks
- It is a test of theory of mind

Your answer is incorrect.

The Sally Anne test is a test of theory of mind whereby children observe a scenario in which a doll (Anne) removes a marble from another doll's (Sally's) basket and places it in a box. The other doll then enters the scene and the children are asked where it will look for its marble. To pass the children must identify that Sally will look in her basket as she does not know that the marble has been moved. Children generally pass it after the age of 4.

The correct answer is: It is a test of theory of mind

Question 61

Not answered

Marked out of 1.00

Siobhan Gibbons is a 72-year-old woman who was admitted to the orthopaedic ward following a fall. On examination she appears confused, she's ataxic and nystagmus is present. You read in her notes that she had a long history of alcohol dependence, and spent many years homeless. During her stay on the ward her confusion becomes less prominent but her cognitive impairment remains. One of the student nurses asks you about Korsakoff Syndrome. Which of the following statements about Korsakoff is not true:

Select one:

- Thiamine acts as a co-factor for alpha-ketoglutatare dehydrogenase
- Reductions in thiamine-dependent enzymes lead to impairment of oxidative metabolism
- Chronic and severe thiamine deficiency result in insufficient concentrations to enable thiamine to cross the blood-brain barrier
- Reductions in thiamine pyrophosphate-dependent enzymes have been found in autopsied cerebellar vermis samples
- Once the features of Korsakoff have been recognised, they can be reversed using IV Thiamine

Your answer is incorrect.

Korsakoff is irreversible; it may be prevented with treatment in the preceding Wernicke's encephalopathy with high dose parenteral thiamine. (Ref: Neuropsychol Rev 2012;22(2):81-92 and Alcohol Clin Exp Res 1993;17(5):1084-1088)

The correct answer is: Once the features of Korsakoff have been recognised, they can be reversed using IV Thiamine

Question 62

Not answered

Marked out of 1.00

Mrs Delia Faber is a 40-year-old woman with a diagnosis of acute intermittent porphyria. Which of the following is FALSE about acute intermittent porphyria?

Select one:

- It is treated with haemin
- Physical signs often include an abdominal rash
- It is an autosomal dominant disorder
- It can be precipitated by use of clonazepam
- Psychiatric side effects often include psychosis

Your answer is incorrect.

Acute Intermittent Porphyria is not typical characterised by a rash; abdominal pain is the most common symptom. It is usually inherited in an autosomal dominant manner but autosomal recessive inheritance has been described. It is caused by mutations in the HMBS gene, which codes for the enzyme porphobilinogen deaminase. It causes acute attacks of abdominal pain and other symptoms, potentially including psychosis. It can be precipitated by the use of medications, including clonazepam, alcohol, low carbohydrate diet, infections and hormone changes during the menstrual cycle. A high-carbohydrate (10% glucose) infusion is recommended, which may aid in recovery. Hematin and heme arginate is the treatment of choice during an acute attack. Heme is not a curative treatment, but can shorten attacks and reduce the intensity of an attack.

The correct answer is: Physical signs often include an abdominal rash

Question 63

Not answered

Marked out of 1.00

Mr Daniel Banner is a 28 year old man who has been brought to A&E by his family as they have been concerned that he has been acting unusually at home and has been isolating himself. Whilst there he informs staff he describes experiencing autoscopic hallucinations. Which of the following is TRUE about this phenomenon?

Select one:

- It describes altered perception of colour
- It is an outcome indicator
- It is seen only in depression
- It is seeing oneself in one's external space
- It is a normal clinical phenomenon

Your answer is incorrect.

An autoscopic hallucination is a visual misperception in which an individual sees himself or herself as if they are an external stimulus.

The correct answer is: It is seeing oneself in one's external space

Question 64

Not answered

Marked out of 1.00

Ron Hyke was subject to a traumatic brain injury following a collision between his car and another vehicle, two-and-a-half years ago. Following a period spent in a coma, he gradually recovered normal functioning in most domains. However, his partner feels that Ron's personality isn't what it used to be. He doesn't have the same humour or interests, he's more predisposed to being emotional, sometimes angry. Which of the following psychological tests would you consider appropriate in testing Ron for post-traumatic personality changes?

Select one:

- The SCID-II
- The Zung depression scale
- The MBTI
- The MMPI-2
- The Portland Adaptability Inventory

Your answer is incorrect.

The Minnesota Multiphasic Personality Inventory (MMPI) is a standardised psychometric test which has been used to test personality change and has some validity in cases of post-traumatic brain injury (see Arch Clin Neuropsychol 2007(22);1:123-130 and 2015(30);1:39-48)

The correct answer is: The MMPI-2

Question 65

Not answered

Marked out of 1.00

Mrs. Brown is attending the memory clinic and undergoing cognitive testing for suspected Alzheimer's dementia. During one part of the examination the psychologist reads a list of 10 words. After 20 minutes and other tasks Mrs. Brown is asked to recall as many words as possible. This is an example of testing

Select one:

- Delayed recognition
- Consolidation
- Semantic memory
- Free recall
- Delayed recall

Your answer is incorrect.

Delayed recall involves the retrieval of previously encoded words, whereas delayed recognition measures only whether or not a word has been encoded and consolidated.

Tests of free recall involve immediate assessment of what is recalled after a list is read e.g. upon hearing the list read, patients are instructed that they will be asked to repeat as many words as they can.

Reading a list of words followed by a 20-minute delay and asking the patient to recall as many words as possible tests 'delayed recall'.

Asking the patient to point out whether a given word was in the original list after a delay is testing 'delayed recognition' as well as consolidation. Semantic memory refers to general world knowledge accumulated through the lifespan.

(Ref: Dierckx E, Engelborghs S, De Raedt R, De Deyn PP, D'Haenens E, Verté D, Ponjaert-Kristoffersen I. The 10-word learning task in the differential diagnosis of early Alzheimer's disease and elderly depression: a cross-sectional pilot study. Aging & mental health. 2011 Jan 1;15(1):113-21)

The correct answer is: Delayed recall

Question 66

Not answered

Marked out of 1.00

According to ICD-10, how long must symptoms be present for before a diagnosis of mania can be made?

Select one:

- 14 days
- 4 days
- 28 days
- 10 days
- 7 days

Your answer is incorrect.

ICD-10 states that manic symptoms must be present for at least 7 consecutive days for a manic episode to be diagnosed. Symptoms must be present for at least 4 days for an episode of hypomania to be diagnosed.

The correct answer is: 7 days

Question 67

Not answered

Marked out of 1.00

Ms Coyle is telling you about the challenges of looking after her three children, alongside managing her long-standing depression. The youngest, Thomas, is 18 months old. She's noticed that when she leaves him on his own to play, he doesn't seem bothered, unlike the other two who used to cry a great deal. When she has visitors to the flat, she finds that Thomas plays alongside them quite calmly, even though he may not have met them before. When she returns to the room after a period of absence, Thomas doesn't seem to notice her much; he rarely makes eye contact. Taking into consideration the experiments conducted by Bowlby and Mary Ainsworth in studying attachment, which attachment style is Ms Coyle describing when telling you about Thomas' behaviour?

Select one:

- Disorganised
- Good enough
- Avoidant
- Anxious ambivalent
- Secure

Your answer is incorrect.

Avoidant attachment is characterised by a desire to maintain emotional distance

The correct answer is: Avoidant

Question 68

Not answered

Marked out of 1.00

Which of the following is a DSM-IV cluster C Personality Disorder?

Select one:

- Histrionic Personality Disorder
- Schizoid Personality Disorder
- Narcissistic Personality Disorder
- Dependent Personality Disorder
- Borderline Personality Disorder

Your answer is incorrect.

The cluster C personality disorders in DSM-IV are dependent, anxious-avoidant and anankastic (obsessive-compulsive).

The correct answer is: Dependent Personality Disorder

Question 69

Not answered

Marked out of 1.00

Susan Miers is a 25-year-old woman with an IQ of 36. How severe is her degree of learning disability?

Select one:

- Severe mental retardation
- Borderline mental retardation
- Mild mental retardation
- Moderate mental retardation
- Profound mental retardation

Your answer is incorrect.

A classification of mild, moderate, severe and profound has been used to describe the degree of learning disability. IQ measurement has traditionally been used to define severity:

• Profound - IQ of less than 20

• Severe - IQ of 20-34

• Moderate - IQ of 35-49

• Mild - IQ of 50-70

The correct answer is: Moderate mental retardation

Question 70

Not answered

Marked out of 1.00

Which of the following Personality Disorders is included in DSM-IV, but NOT ICD-10?

Select one:

- Histrionic Personality Disorder
- Dependent Personality Disorder
- Schizotypal Personality Disorder
- Schizoid Personality Disorder
- Paranoid Personality Disorder

Your answer is incorrect.

The World Health Organization's ICD-10 uses the name schizotypal disorder. It is classified as a clinical disorder associated with schizophrenia, rather than a personality disorder as in DSM-5.

The correct answer is: Schizotypal Personality Disorder

Question 71

Not answered

Marked out of 1.00

Graham has asked for a consultation to discuss his concerns regarding his partner Sharon. He's read about histrionic personality disorder, and wonders whether Sharon's struggles may be in keeping with this diagnosis. He describes her to you. Which of the following descriptions is not part of the criteria for a diagnosis of histrionic personality disorder?

Select one:

- Her emotions tend to fluctuate rapidly, and others say that she can come across as quite shallow
- She often wears revealing sexually provocative clothing
- She appears not to mind whether she is praised or criticised
- She is easily influenced by others
- At a party, she always talks loudly and has to be the centre of attention

Your answer is incorrect.

People with histrionic personality disorder are very sensitive to criticism, and often seeking out approval and reassurance.

ICD
F60.4 Histrionic personality disorder
Personality disorder characterized by:
(a)self-dramatization, theatricality, exaggerated expression of emotions;
(b)suggestibility, easily influenced by others or by circumstances;
(c)shallow and labile affectivity;
(d)continual seeking for excitement and activities in which the patient is the centre of attention;
(e)inappropriate seductiveness in appearance or behaviour;
(f)over-concern with physical attractiveness.

Associated features may include egocentricity, self-indulgence, continuous longing for appreciation, feelings that are easily hurt, and persistent manipulative behaviour to achieve own needs.

The correct answer is: She appears not to mind whether she is praised or criticised

Question 72

Not answered

Marked out of 1.00

Which of the following is true according to both the ICD-10 and DSM-IV descriptions of schizophrenia?

Select one:

- Undifferentiated schizophrenia is not described
- Catatonic schizophrenia is not described
- Residual schizophrenia is no longer a feature
- Paranoid schizophrenia is not a feature
- Simple schizophrenia has been removed

Your answer is incorrect.

Simple schizophrenia was not included amongst the schizophrenia subtypes in DSM-IV. In DSM-V, all schizophrenia subtypes have now been removed.

The correct answer is: Simple schizophrenia has been removed

Question 73

Not answered

Marked out of 1.00

Paola Pozzini is a 74-year-old woman, who was widowed seven years previously. She has been calling the police in order to report her neighbours, who she believes are transmitting cannabis from their flat into her water system, so that the water that emerges out of her taps smells of drugs. She believes they wish to pressure her to move out of her flat, so they can take it over and convert it into a drugs den. This is making Mrs Pozzini feel very anxious and agitated. She holds her beliefs absolutely. Which of the following criteria, according to the ICD-10, is necessary for a diagnosis of persistent delusional disorder?

Select one:

- It must include a mood incongruent component
- It cannot be diagnosed if the patient is living with someone else
- The delusions must be present for at least one month
- It can be diagnosed alongside paranoid schizophrenia
- It is accompanied by a dementing illness

Your answer is incorrect.

All the others are not included in the ICD-10. The delusions are usually mood congruent. The diagnosis cannot be made alongside a diagnosis of paranoid schizophrenia or if accompanied by a dementing illness.

The correct answer is: The delusions must be present for at least one month

Question 74

Not answered

Marked out of 1.00

Latah is a culturally specific syndrome occurring amongst people from Southeast Asia. Under which of the following is it classified in ICD-10?

Select one:

- Adjustment disorder
- Psychotic disorder
- Dissociative disorder
- Other neurotic disorder
- Anxiety disorder

Your answer is incorrect.

Latah is classified under other specified neurotic disorders, along with many other culturally specific syndromes.

F48.8 Other specified neurotic disorders:
Includes:
Briquet's disorder
Dhat syndrome
koro
latah
occupational neurosis, including writer's cramp
psychasthenia
psychasthenic neurosis
psychogenic syncope

The correct answer is: Other neurotic disorder

Question 75

Not answered

Marked out of 1.00

In the context of a loss event and subsequent grief, the term incentive salience refers to which of the following?

Select one:

- Lack of rewarding sensation from pleasurable activities
- Reduced interest in items that remind us of the dead
- Attentional bias towards the objects that belonged to the dead person
- Reality distortion in the presence of cues related to the dead person
- Cognitive bias towards the benefits of a death event

Your answer is incorrect.

Why should a loss event make us feel sad universally? One explanation is offered through the model of incentive salience for grief (Freed and Mann, 2007). When an attachment figure dies (or leaves us), the rewarding feeling that we generally associate with the cues associated with them persists (e.g. photographs, the daily objects that they used etc.). These cues, given their associated incentive salience, automatically bias our attention towards them and trigger the unpleasant subjective yearning and sense of loss. This excessive attention toward an unavailable attachment figure serves to maintain grief's core symptoms. Successfully reducing this elevated incentive salience during separation will help one to accept the loss. Sadness that emerges during grief reaction is an adaptive coping response, which serves to replace the association of rewarding, incentive salience with an unpleasant feeling that eventually serves to produce detachment. Ref: Freed PJ, Mann JJ. Sadness and loss: Toward a neurobiopsychosocial model. American Journal of Psychiatry. 2007 Jan;164(1):28-34.

The correct answer is: Attentional bias towards the objects that belonged to the dead person

Question 76

Not answered

Marked out of 1.00

You are interviewing Dinah Mensah, a 37-year-old woman with a diagnosis of schizophrenia, who was admitted to the ward seven months previously. You are finding her speech to be unusual. Here's an example of her communication: 'when I went to the forest I was there by the herbal herbal bal bal balbalbal plot of land in that region, near the mountain tain tain tain tain....'. What word best describes her speech?

Select one:

- Perseveration
- Flight of ideas
- Derailment
- Echolalia
- Logoclonia

Your answer is incorrect.

In logoclonia there is meaningless repetition of the last syllables of a word or phrase.

The correct answer is: Logoclonia

Question 77

Not answered

Marked out of 1.00

Mr. Terry Charlesworth is a 43-year-old man with a long-standing history of schizophrenia. He attends clinic to see his psychiatrist. When the doctor offers to shake hands, he takes his hand 20 times but does not shake it. Which of the following best describes this phenomenon?

Select one:

- Ambitendency
- Mitgehen
- Waxy flexibility
- Posturing
- Negativism

Your answer is incorrect.

Ambitendency is a phenomenon in which an individual alternates between resistance to and cooperation with instructions or prompts.

In negativism a patient resists attempts to move them and does the opposite of what is asked, seemingly without any rationale.

In Mitgehen very slight pressure on a part of the patient's body leads to movement in any direction, also called the "anglepoise" effect or "anglepoise lamp sign". This is done despite instructions that the patient resist the pressure. A less extreme version is termed mitmachen.

Posturing is a spontaneous and active maintenance of a posture against gravity, seemingly without purpose.

Waxy flexibility (cerea flexibilitas) is characterized by a patient's movements having the feeling of a plastic resistance, as if the person were made of wax.

The correct answer is: Ambitendency

Question 78

Not answered

Marked out of 1.00

Mr. James O'Brien is a 25-year-old man with a diagnosis of schizophrenia. He has experienced some voices and persecutory delusions but his main difficulties arise from the negative symptoms associated with his illness. Which of the following are negative symptoms of schizophrenia?

Select one:

- Diminished sleep
- Emotional insensitivity
- Elective mutism
- Apathy
- Waxy flexibility

Your answer is incorrect.

The negative symptoms of schizophrenia are: apathy, blunted or absent emotional responses, reduced speech, social withdrawal, diminished attention, anhedonia, sexual problems, lethargy.

The correct answer is: Apathy

Question 79

Not answered

Marked out of 1.00

Various forms of formal thought disorder have been described which each having distinct characteristics. Which of the following are characterised by a lack of causal links in the content of speech?

Select one:

- Metonymy
- Derailment
- Drivelling
- Asyndesis
- Fusion

Your answer is incorrect.

Asyndesis describes a lack of genuine causal links in which clusters of more or less related sequences of thoughts, instead of well-knit sequences of thoughts are used, creating a feeling of vagueness.

Derailment is a form of thought disorder characterised by a sudden change in the train of thought and/or speech, either spontaneously or in response to an internal stimulus, from the topic's track onto another which is obliquely related or entirely unrelated.

Metonymy involves approximate uses of words, such as paperskate rather than pen.

Fusion involves thoughts being fused together, leading to a loss of goal direction.

In **drivelling** there is a disordered intermixture of the constituent parts of one complex thought.

The correct answer is: Asyndesis

Question 80

Not answered

Marked out of 1.00

Which of the following best describes the sensation or concept of interoception?

Select one:

- Awareness of hallucinations
- Awareness of one's physiological state
- Delusional perception of internal organs
- Reflecting on one's behaviours
- Location of joints

Your answer is incorrect.

Interoception is defined as the sense of the internal state of the body. It encompasses the brain's process of integrating signals relayed from the allowing for a nuanced representation of the physiological state of the body. This is important for maintaining homeostatic conditions in the body and potentially aiding in self-awareness. It is thought to be impaired in people with ASD.

The correct answer is: Awareness of one's physiological state

Question 81

Not answered

Marked out of 1.00

Mr. Greg Fallon is a 45-year-old man who has been taken to the GP by his family, as they have been worried about him. He has been insisting that he worked at Buckingham Palace many years ago, despite having not done so and has telling people at work about it. Which of the following best describes this phenomenon?

Select one:

- Delusional memory
- False memory
- Retrospective falsification
- Reduplicative paramnesia
- Confabulation

Your answer is incorrect.

This is most likely to be a delusional memory. This is the symptom when the patient recalls as remembered an event or idea that is clearly delusional in nature. The delusion is retrojected in time. These are sometimes called retrospective delusions.

Ref: McAllister-Williams RH. The description of primary delusions: confusion in standard texts and among clinicians. Psychiatric Bulletin. 1997 Jun;21(6):346-9.

The correct answer is: Delusional memory

Question 82

Not answered

Marked out of 1.00

Mrs. Tracey Powers is a 42-year-old woman who describes being able to recall daily events in great detail, "like watching a video". Which of the following describe this form of memory?

Select one:

- Echoic
- Eidetic
- Iconic
- Procedural
- Flashbulb

Your answer is incorrect.

This is most likely to represent eidetic memory, also called photographic memory. It is unclear if this is a real phenomenon.

The correct answer is: Eidetic

Question 83

Not answered

Marked out of 1.00

Which of the following types of formal thought disorder is characterised by two or more thoughts being interwoven into one?

Select one:

- Derailment
- Neologism
- Alogia
- Fusion
- Drivelling

Your answer is incorrect.

Fusion involves thoughts being fused together, leading to a loss of goal direction.

Alogia involves a lack of unstimulated thought content, usually seen as a lack of speech.

In **drivelling** there is a disordered intermixture of the constituent parts of one complex thought.

A **neologism** describes an invented word or a known word being used to represent an under-related concept, e.g. sausage for car.

Derailment is a form of thought disorder characterised by a sudden change in the train of thought and/or speech, either spontaneously or in response to an internal stimulus, from the topic's track onto another which is obliquely related or entirely unrelated.

The correct answer is: Fusion

Question 84

Not answered

Marked out of 1.00

Miss Anne Barlow is a 30-year-old woman who has a history of episodes of catatonic schizophrenia. She has recently been readmitted after experiencing a relapse with symptoms typical of her previous episodes of illness. Which of the following is a ICD-10 criteria for catatonic schizophrenia?

Select one:

- Mutism
- Waxy flexibility
- Excitability
- Reduced muscular tone
- Mannerisms

Your answer is incorrect.

The criteria for catatonic schizophrenia are:

a. Stupor (marked decrease in reactivity to the environment and in spontaneous movements and activity) or mutism;

b. Excitement (apparently purposeless motor activity, not influenced by external stimuli);

c. Posturing (voluntary assumption and maintenance of inappropriate or bizarre postures);

d. Negativism (an apparently motiveless resistance to all instructions or attempts to be moved, or movement in the opposite direction);

e. Rigidity (maintenance of a rigid posture against efforts to be moved);

f. Waxy flexibility (maintenance of limbs and body in externally imposed positions);

g. Other symptoms such as command automatism (automatic compliance with instructions), and perseveration of words and phrases.

The correct answer is: Waxy flexibility

Question 85

Not answered

Marked out of 1.00

Which of the following is a method for teaching certain behaviours to children with intellectual disability in which each step gives a cue for the next action to be performed?

Select one:

- Chaining
- Prompting
- Rewinding
- Shaping
- Cueing

Your answer is incorrect.

Chaining involves breaking a complex behaviour into steps and then teaching one step at a time, adding a new one to those that have come before. The previous steps act as a prompt for the new one, thus building a chain.

Shaping involves giving positive reinforcement for approximations of a behaviour until it is mastering, e.g. congratulating a child for holding a fork when they hold it by the tines rather than the handle when teaching them to use cutlery.

Prompting involves reminding a child of what they need to be doing, however this is done.

In the context of memory retrieval, **cueing** is achieved via prompts or cues, which include anything that is connected in some way to target information in long-term memory, prompting retrieval of it.

The correct answer is: Chaining

Question 86

Not answered

Marked out of 1.00

Gestalt psychology describes 5 distinct principles of grouping. Which of the following is one of these principles?

Select one:

- Irritability
- Openness
- The law of distinct fates
- Separation
- Proximity

Your answer is incorrect.

The 5 principles of Gestalt psychology are: proximity, similarity, good continuation, closure, and the law of common fate. They were described to explain human perception of groups of objects and how we perceive parts of objects and form whole objects on the basis of these. The concept of proximity is sometimes referred to as the principle of distance.

The correct answer is: Proximity

Question 87

Not answered

Marked out of 1.00

Counter-conditioning is a psychological technique used in which of the following psychotherapeutic interventions?

Select one:

- Rational emotive therapy
- Systematic desensitisation
- Psycho-education
- Psychodynamic psychotherapy
- Cognitive behavioural therapy

Your answer is incorrect.

Counter-conditioning is a technique that is often employed as a part of systematic desensitisation when managing phobias and similar disorders. It involves the conditioning of an unwanted behaviour or response to a stimulus into a wanted behaviour or response by the association of positive actions with the stimulus. For example, when managing a phobia of dogs, a therapist may aim to associate an anxiety-provoking stimulus, such as a dog being in the next room, with a positive reward. Therefore, this will associate the positive reward with the stimulus and diminish the anxiety associated with the latter.

Wolpe, the proponent of systematic desensitization used Guthrie's (1952) concept of counterconditioning to explain the long-term benefits of systematic desensitisation and claimed that desensitization replaced anxiety with relaxation.

(Ref: Tryon, WW., in Cognitive Neuroscience and Psychotherapy, 2014)

The correct answer is: Systematic desensitisation

Question 88

Not answered

Marked out of 1.00

Sergio Cacin and his family are preparing to travel to South America for a family holiday. Sergio has struggled with a number of phobias to particular situations for which he's engaged in some behavioural therapy (relaxation training, exposure and response prevention) with a good outcome. His fear of snakes has subsided, but he discovers that other members of his family, even though they've not struggled with anxiety to the extent he has, also seem to have an aversive reaction to the sight of snakes. What theory can explain this phenomenon?

Select one:

- Habituation
- Avoidance
- Stimulus preparedness
- Generalisation
- Extinction

Your answer is incorrect.

Stimulus preparedness implies that fear-relevant stimuli are biologically prepared, as opposed to only occurring in those who already fear snakes.

The correct answer is: Stimulus preparedness

Question 89

Not answered

Marked out of 1.00

Which of the following best describes the Hawthorne Effect?

Select one:

- Academic performance based on the expectation of others
- Belief that an individual's behaviour is representative of the behaviour of a group
- Temporary increase in work performance due to the presence of others
- Tendency to perceive someone as wholly bad based on a small number of attributes
- Not helping someone in need due to the presence of others

Your answer is incorrect.

The **Hawthorne** effect describes the phenomenon of individuals or a group changing their behaviour in response to being observed.

The **bystander** effect is a social psychological phenomenon in which individuals are less likely to offer help when other people are present.

The **horn** effect, closely related to the **halo** effect, is a form of cognitive bias that causes one's perception of another to be unduly influenced by a single negative trait.

The **Pygmalion** effect, or **Rosenthal** effect, is a psychological phenomenon wherein high expectations lead to improved performance in a given area. The targets of the expectations internalise their positive labels, and those with positive labels succeed accordingly; a similar process works in the opposite direction in the case of low expectations.

The **group attribution error** refers to people's tendency to believe either (1) that the characteristics of an individual group member are reflective of the group as a whole, or (2) that a group's decision outcome must reflect the preferences of individual group members, even when external information is available suggesting otherwise.

The correct answer is: Temporary increase in work performance due to the presence of others

Question 90

Not answered

Marked out of 1.00

John Guilmot is a 75-year-old man who has been subject to a cerebro-vascular event affecting his medial parietal lobe. You wish to test some of his cognitive capacities. You present him with a list of words and ask him to read them through. You then show him a second list and ask him to discern which of the words he sees are on both lists. What aspect of cognitive functioning are you testing?

Select one:

- Recognition
- Cued recall
- Free recall
- Serial recall
- Declaration

Your answer is incorrect.

The scenario describes one of the simplest forms of testing for recognition, which makes the retrieval process easy to record and analyse.

(Ref: Radvansky (2006) Human Memory. Boston PEG).

The correct answer is: Recognition

Question 91

Not answered

Marked out of 1.00

Abraham was treated with ECT following his depressive episode. Which of the following persists after ECT?

Select one:

- Disorientation to time
- General confusion
- Retrograde amnesia
- Disorientation to place
- Anterograde amnesia

Your answer is incorrect.

Retrograde amnesia, particularly for autobiographical memory is the most persistent complication after ECT. Confusion and disorientation usually resolves within 24-48 hours. Anterograde amnesia usually resolves within few weeks.

The correct answer is: Retrograde amnesia

Question 92

Not answered

Marked out of 1.00

This type of aggression is often planned and may be used to attain a goal

Select one:

- Overt aggression
- Instrumental aggression
- Positive aggression
- Hostile aggression
- Covert aggression

Your answer is incorrect.

Instrumental aggression is used with a specific goal in mind, such as obtaining money in the course of an armed robbery.

Hostile aggression is in response to a stimulus or provocation.

Aggression can be overt and easily identifiable, or covert and "passive-aggressive", e.g. bullying.

The correct answer is: Instrumental aggression

Question 93

Not answered

Marked out of 1.00

Which of the following is true regarding self-serving bias?

Select one:

- It is a type of attribution error
- It occurs in depression
- It attributes internal failures to internal sources
- It is a form of self-compassion
- It attributes external success to external sources

Your answer is incorrect.

Some people attribute their successes to internal factors and their failures to external factors. It is called the self-serving bias, which helps to boost self-esteem.

The correct answer is: It is a type of attribution error

Question 94

Not answered

Marked out of 1.00

The tendency for a group to make more risky decisions than individuals otherwise would make is known as

Select one:

- Conformity
- Group polarisation
- Risky shift
- Obedience
- Groupthink

Your answer is incorrect.

Risky shift describes the phenomenon whereby a group will make more risky decisions than an individual.

This is similar to group polarisation in which a group makes more extreme decisions than the individuals would, but in terms of their initial inclination, either cautious or risky.

Groupthink involves putting the harmony of the group foremost and avoiding disagreement at all costs, leading to an irrational decision-making process.

Conformity is the act of matching attitudes, beliefs, and behaviors to group norms.

Obedience involves a person yielding to explicit instructions or orders from an authority.

The correct answer is: Risky shift

Question 95

Not answered

Marked out of 1.00

Wynne Flanders is a 17-year-old who recently sat her school exams. For the past three months, following the separation of her parents, she's been feeling demotivated, finding it difficult to concentrate on her studies, feeling lethargic, sleeping more than usual. With her close friend Tamara, she goes to school to pick up their exam results. They both get a C as their final result. Tamara says that the reason she got a C is because their teacher was useless and didn't explain the concepts required for the exam clearly enough; she's sure that based on her own capacities, had she been properly coached, she would have achieved an A grade. Wynne on the other hand, isn't surprised by her grade, and thinks that it's her own incompetence that meant she didn't get a higher grade. This scenario illustrates the concept of:

Select one:

- Hawthorne effect
- Self-serving bias
- Anchoring bias
- Halo effect
- Selection bias

Your answer is incorrect.

Self-serving bias is a type of cognitive bias in which positive events are attributed to one's own character, and negative events to external factors. This can be reversed in depression.

The correct answer is: Self-serving bias

Question 96

Not answered

Marked out of 1.00

Minority influence is a form of social influence whereby an individual or minority group persuades the majority to accept their beliefs or behaviour. Which of the following increases minority influence?

Select one:

- Being consistent
- Not being flexible
- Disregarding the majority view
- Emphasising their difference from the majority
- Hostility to majority figureheads

Your answer is incorrect.

Minority influence is increased by consistency in messaging, flexibility, appealing to the majority view and persuading majority figureheads to help propagate their views. If the minority is seen as being more similar to the majority, it also increases their influence.

Ref: Nemeth CJ. Minority influence theory. Handbook of theories of social psychology. 2011 Aug 3;2:362-378.

The correct answer is: Being consistent

Question 97

Not answered

Marked out of 1.00

When there is a discrepancy between co-existing but contradictory beliefs or attitudes, which of the following is an individual most likely to do?

Select one:

- Seek information about the dissonant belief
- Change one of their beliefs or attitudes
- Emphasise the importance of the contrast
- Weigh the available information objectively
- Discuss their difficulty with others

Your answer is incorrect.

Individuals experiencing cognitive dissonance can resolve this by:

- Changing behaviours
- Changing beliefs or attitudes to reduce the dissonance
- Reduce the importance of the dissonance
- Ignore information that emphasises the dissonance

Of these, changing behaviours is the least likely to occur. Individuals do not tend to acknowledge the dissonance explicitly as this would involve recognising that their thoughts are contradictory, which is psychologically threatening. They are most likely to change their beliefs or attitudes to reduce the dissonance.

The correct answer is: Change one of their beliefs or attitudes

Question 98

Not answered

Marked out of 1.00

There is evidence from sociopsychological studies that when individuals believe that someone else can respond to a situation, they are less likely to respond. Which of the following describes this phenomenon?

Select one:

- Dissolution of responsibility
- Division of labour
- Diffusion of responsibility
- Groupthink
- Social loafing

Your answer is incorrect.

Diffusion of responsibility describes the phenomenon of a person being less likely to take responsibility for action or inaction when others are present. It is considered a form of attribution as the individual assumes that others either are responsible for taking action or have already done so.

The correct answer is: Diffusion of responsibility

Question 99

Not answered

Marked out of 1.00

Asch's studies of group dynamics focused around conformity and integration. According to his theories, which of the following would lead to disintegration of a group?

Select one:

- Younger age
- Equality between members
- Larger group size
- Difficult work
- More public interactions

Your answer is incorrect.

Asch theorised that group conformity was more likely with larger numbers, more challenging tasks, conversations being more public and individuals perceiving other members as being more senior than them.

Ref: Hodges, B. H., and Geyer, A. (2006). A nonconformist account of the Asch experiments: values, pragmatics, and moral dilemmas. Pers. Soc. Psychol. Rev. 10, 2-19.

The correct answer is: Equality between members

Question 100

Not answered

Marked out of 1.00

Social pressure within groups can have a significant impact on the attitudes and behaviours of an individual. Which of the following would increase the likelihood of an individual expressing ambivalence?

Select one:

- The presence of a charismatic leader
- There is a high degree of agreement in the group
- Need to promote harmony within the group
- Feels less identified with group
- Need to appear as part of the group

Your answer is incorrect.

A lesser degree of identification with a given group increases the likelihood an individual will express ambivalence or a dissenting opinion.

Ref: Brickson S. The impact of identity orientation on individual and organizational outcomes in demographically diverse settings. Academy of management Review. 2000 Jan 1;25(1):82-101.

The correct answer is: Feels less identified with group

Question 101

Not answered

Marked out of 1.00

Jennifer Zane is a 12-year-old girl who is increasingly influenced by her peers and generally tries to act in a way that others will approve of. According to Kohlberg's theory of moral development, which stage is she going through?

Select one:

- Social contract orientation
- Interpersonal accord and conformity
- Self-interest orientation
- Obedience and punishment orientation
- Authority and social-order maintaining orientation

Your answer is incorrect.

She is likely in the conventional stage of interpersonal accord and conformity, or the good boy/girl orientation whereby individuals conform to social standards and are receptive to approval or disapproval from others as it reflects society's views. They try to be a "good boy" or "good girl" to live up to these expectations having learned that being regarded as good benefits the self.

The correct answer is: Interpersonal accord and conformity

Question 102

Not answered

Marked out of 1.00

Which of the following is true about ethnic differences in access to mental health services in the UK?

Select one:

- Cultural competence of the providers does not affect access to care
- Most ethnic minorities that receive mental health care do so via their general practitioners
- Psychotic disorders are less often diagnosed among black men
- Among the different ethnic groups, anxiety and depression are highest among white women
- Black and South Asian individuals are less likely to obtain mental healthcare

Your answer is incorrect.

Among BME communities, mental illnesses are more prevalent but less treated.

There is relatively higher prevalence of anxiety and depression in South Asian women (63.5% compared with 28.5% of white women), and psychotic disorders in Afro-Caribbean men (3.1% compared with 0.2% of white men).

In general, people from ethnic minorities are less likely than their White British counterparts to have contacted their GP about mental health issues in the last year, to be prescribed antidepressants, or to be referred to specialist mental health services. There is some evidence that hard-to-reach groups, including BME, are less likely to obtain appropriate mental healthcare and may be disadvantaged either because they are unable to access services or because they receive inadequate help from services.

Factors considered important barriers to accessing services include cultural competence (i.e. the ability of providers and organisations to effectively deliver healthcare services that meet the social, cultural and linguistic needs of patients); cultural models of illness; stigma; fear and perceived discrimination and barriers to communication.

Ref: Excerpt taken from Memon A, Taylor K, et al. BMJ open. 2016 Nov 1;6(11):e012337.

The correct answer is: Black and South Asian individuals are less likely to obtain mental healthcare

Question 103

Not answered

Marked out of 1.00

Mr Daio is a 35-year-old Senegalese man who is brought to A&E by his family after becoming suddenly aggressive to his wife. He accused her of being a witch and tried to exorcise demons from her. He claimed to have seen the demons and became agitated and distressed. Which of the following culture-bound syndromes best fits this description?

Select one:

- Bouffée délirante
- Latah
- Ataque de nervios
- Koro
- Brain fag

Your answer is incorrect.

Bouffée délirante describes sudden attacks of brief duration with paranoid delusions and often concomitant hallucinations, typically precipitated by an intense fear of magical persecution through sorcery or witchcraft. It is a syndrome occurring in Francophone West Africa and Haiti.

Ataque de nervios translates as "attack of nerves" and is commonly associated with Hispanic populations, with symptoms including uncontrollable screaming or shouting, crying, trembling, sensations of heat rising in the chest and head, dissociative experiences, and verbal or physical aggression. The reaction is usually associated with a stressful event relating to the family.

Koro is a culture-specific syndrome delusional disorder in which an individual has an overpowering belief that one's genitalia are retracting and will disappear, despite the lack of any true longstanding changes to the genitals. It is more associated with Asian populations but there have been epidemics in Africa, the US and Europe.

Latah is most commonly seen in Southeast Asia and is characterised by abnormal behaviors result from a person experiencing a sudden shock. The affected person typically engages in such behaviours as screaming, cursing, dancing type movements, and uncontrollable laughter, and will typically mimic the words or actions of those around them.

Brain fag is commonly seen in West African students and is characterised by symptoms including somatic, sleep-related and cognitive complaints, head and neck pains, difficulty in concentrating and retaining information, and eye pain. It is thought to be related to stress from studying.

The correct answer is: Bouffée délirante

Question 104

Not answered

Marked out of 1.00

Which of the following terms is used to describe social disadvantage as a result of disability?

Select one:

- Handicap
- Felt stigma
- Enacted stigma
- Sickness
- Illness

Your answer is incorrect.

Handicap is the social disadvantage resulting from disability.

Enacted stigma (external stigma, discrimination) refers to the experience of unfair treatment by others.

Felt stigma (internal stigma or self-stigmatization) refers to the shame and expectation of discrimination that prevents people from talking about their experiences and stops them seeking help.

The correct answer is: Handicap

Question 105

Not answered

Marked out of 1.00

Which of the following is FALSE about the concept of high expressed emotion?

Select one:

- Patients with schizophrenia whose families have high expressed emotion are at an increased risk of relapse
- In patients with schizophrenia whose families have high expressed emotion, males have a higher risk of relapse than females
- The domains of critical comments, emotional over-involvement and hostility were most predictive of relapse
- The proportion of high expressed emotion in carers of patients with schizophrenia is over 50%
- Expressed emotions are measured using the Camberwell Family Interview

Your answer is incorrect.

There is no evidence for a difference in the rate of relapse for men and women with schizophrenia exposed to high expressed emotion.

The correct answer is: In patients with schizophrenia whose families have high expressed emotion, males have a higher risk of relapse than females

Question 106

Not answered

Marked out of 1.00

Mr Janzeb Khan is a 33-year-old Kashmiri man who works in an office in London. He speaks fluent English and often goes to the pub with his colleagues after work. At home, he speaks fluent Kashmiri and attends mosque with his family. In terms of culture he can be said to be:

Select one:

- Merged
- Assimilated
- Separated
- Multi-culturalised
- Integrated

Your answer is incorrect.

Cultural integration is a form of cultural exchange in which one group assumes the beliefs, practices and rituals of another group without sacrificing the characteristics of its own culture.

Cultural assimilation occurs when members of a minority group become indistinguishable from the dominant group.

Acculturation refers to the socialisation process by which foreign-born individuals adopt the values, customs, norms, attitudes, and behaviours of the dominant host culture.

Enculturation is the process by which people learn the dynamics of their surrounding culture and acquire values and norms appropriate or necessary in that culture and worldviews.

The correct answer is: Integrated

Question 107

Not answered

Marked out of 1.00

Which of the following is not true about Kohlberg's theory of moral development?

Select one:

- Individuals can display features of more than one stage at a time
- All adults reach the post-conventional stage
- Individuals cannot skip stages
- The stages are not culturally neutral
- Individuals in conventional stages compare their actions with views of society

Your answer is incorrect.

All adults do not reach the post-conventional stages of moral development. At the same time, people in different cultures seem to move through the sequence at different rates and to reach different end-points. Edwards et al., in 1980, reported that in the West, most urban middle-class adults reach stage 4 (conventional), with a small percentage using some stage 5 (post-conventional) reasoning. In the isolated villages and tribal communities of many countries, however, it is rare to find any adult beyond stage 3.

Ref: Kohlberg L. Stages of moral development. Moral education. 1971;1(51):23-92 and Edwards CP. Cross Cultural Research on Kohlberg's Stages: The Basis for Consensus.

The correct answer is: All adults reach the post-conventional stage

Question 108

Not answered

Marked out of 1.00

Miss Victoria Pierce is a 20-year-old woman with a diagnosis of anorexia nervosa who has been referred for a psychological assessment. Which of the following defence mechanisms is she most likely to utilise?

Select one:

- Fantasy, avoidance
- Splitting, acting out
- Denial, rationalization
- Denial, projection
- Projection, splitting

Your answer is incorrect.

Denial, rationalisation and intellectualisation are commonly seen in patients with anorexia nervosa.

The correct answer is: Denial, rationalization

Question 109

Not answered

Marked out of 1.00

Mr Davies is a 78-year-old man who has brought his wife to the GP because he is concerned about some paranoid ideas she has been having. He matter-of-factly tells the GP about why he thinks she is behaving like this, and where the ideas might have come from. The defence mechanism he is utilizing is:

Select one:

- Isolation
- Repression
- Suppression
- Intellectualisation
- Rationalisation

Your answer is incorrect.

Intellectualisation is a form of defence mechanism where an individual concentrates on the intellectual components of a situation so as to distance oneself from the associated anxiety-provoking emotions.

Rationalisation is a defence mechanism where the individual devises reassuring or self-serving but *incorrect* explanations for his or her own or others' thoughts, actions, or feelings.

The correct answer is: Intellectualisation

Question 110

Not answered

Marked out of 1.00

Which of the following is NOT a Kleinian defence?

Select one:

- Isolation
- Projective identification
- Splitting
- Introjection
- Denial

Your answer is incorrect.

Isolation is not a Kleinian defence. It is characterised as a mental process involving the creation of a gap between an unpleasant or threatening cognition, and other thoughts and feelings. By minimising associative connections with other thoughts, the threatening cognition is remembered less often and is less likely to affect self-esteem or the self-concept. The other defences listed are more primitive and were addressed more by Klein in her work with children.

The correct answer is: Isolation

Question 111

Not answered

Marked out of 1.00

Which of the following defences is known as the Primary Defence?

Select one:

- Splitting
- Repression
- Denial
- Isolation
- Suppression

Your answer is incorrect.

Repression has been referred to the Primary Defence for the unconscious as it involves preventing thoughts or impulses becoming conscious.

The correct answer is: Repression

Question 112

Not answered

Marked out of 1.00

Which of the following is a mature defence?

Select one:

- Intellectualisation
- Suppression
- Splitting
- Somatisation
- Acting out

Your answer is incorrect.

Suppression is the conscious decision to delay paying attention to a thought, emotion, or need in order to cope with the present reality; making it possible later to access uncomfortable or distressing emotions whilst accepting them.

The correct answer is: Suppression

Question 113

Not answered

Marked out of 1.00

According to Freud's dual instinct theory, which of the following instincts is the dominant instinct in human psychology?

Select one:

- Thanatos
- Archetypes
- Animus
- Survival
- Comfort

Your answer is incorrect.

Dual instinct theory looks at Eros and Thanatos as life and death instincts. Freud believed that Thanatos underlay and accompanied Eros and so could be seen as more primal.

The correct answer is: Thanatos

Question 114

Not answered

Marked out of 1.00

Sam Allen is a 15-year-old boy who has had an argument with his mother about mess in his room and later cleans it without being asked. Which of the following defences is he most likely to be utilising?

Select one:

- Repression
- Reaction formation
- Undoing
- Altruism
- Sublimation

Your answer is incorrect.

In undoing an individual tries to “undo" an unhealthy, destructive or otherwise threatening thought by acting out the reverse of the unacceptable. It involves symbolically nullifying an unacceptable or guilt provoking thought, idea, or feeling by confession or atonement.

The correct answer is: Undoing

Question 115

Not answered

Marked out of 1.00

Some animals demonstrate the phenomenon of perceiving the first moving object they see after birth as being their mother. Which of the following best describes this?

Select one:

- Classical conditioning
- Stimulus generalisation
- Imputing
- Imprinting
- Westermark effect

Your answer is incorrect.

This describes filial **imprinting** that is often seen in birds. Newly hatched chicks will identify an object seen after birth as their mother and follow it persistently. This tends to occur within a certain critical time period shortly after hatching. More generally, imprinting is any kind of phase-sensitive learning (learning occurring at a particular age or a particular life stage) that is rapid and apparently independent of the consequences of behaviour.

The Westermark effect is seen in instances where two people who live in domestic proximity during the first few years in the life of either one become desensitised to later close sexual attraction.

Classical conditioning refers to a learning procedure in which a biologically potent stimulus (e.g. food) is paired with a previously neutral stimulus (e.g. a bell).

In classical conditioning, **stimulus generalisation** is said to occur if, after a particular conditioned stimulus has come to elicit a response, another similar stimulus will elicit the same response.

The correct answer is: Imprinting

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